

Title 14—AERONAUTICS AND SPACE

Chapter I—Federal Aviation Adminis- tration, Department of Transportation

[Docket No. 10129; Amdts. Nos. 23-11; 25-27;
27-6; 29-7; and 91-90]

ANTICOLLISION LIGHT STANDARDS

The purpose of these amendments is to (1) permit the use of either aviation red or aviation white anticollision light systems; (2) expand the chromaticity-coordinate range for aviation white; (3) increase the minimum effective intensities for anticollision lights installed on all aircraft to be type certificated after the effective date of these amendments; and (4) require that within 1 year after the effective date of these amendments all powered civil aircraft with standard airworthiness certificates have an approved anticollision light system for night flight.

These amendments are based on a notice of proposed rule making (Notice 70-21) published in the *FEDERAL REGISTER* on June 4, 1970 (35 F.R. 8665). Numerous comments were received in response to the notice and except to the extent that a comment merely repeated issues which were discussed in detail and properly disposed of in the notice, the FAA's disposition of those comments are set forth hereinafter.

Several comments objected to the requirement that all aircraft flying at night must be fitted with an approved anticollision light. One commentator pointed out that certain operators of small aircraft may fly for a few minutes after dark but only to reach a destination and that such operators should not be burdened with an expensive anticollision light requirement. Another commentator took a similar position with respect to aircraft that are operated at night only in local areas or only in an emergency. The FAA does not agree. With proper flight planning, flight after dark can usually be avoided without inconvenience. Moreover, any aircraft operating at night without an approved anticollision light constitutes a hazard regardless of whether it is being operated in a local area or in an emergency.

One commentator indicated that the FAA may have underestimated the problems associated with removal of the red filter on the anticollision light on existing aircraft. The commentator pointed out that it may be impossible to meet the current field-of-coverage requirements because of the need to mask the unfiltered light to eliminate back-scatter. While the FAA recognizes this problem, it should be noted that it was not proposed to require anyone to remove the

red filters on existing aircraft. Moreover, if an operator elected to do so, he would not be required to meet the new intensity requirements. Furthermore, he could relocate the light to minimize the back-scatter problem.

There were numerous comments received in response to Notice 70-21 which recommended changes to the regulations in addition to those proposed in the notice. In this connection, it was recommended that the current field-of-coverage requirements be extended; that the existing flash-frequency requirements be modernized; that exterior minimum intensity requirements be increased beyond the values proposed in the notice; that a red anticollision light be required for nighttime use and a white anticollision light be required for daytime use; and that the operating rule in Part 91 also cover aircraft other than powered civil aircraft with standard airworthiness certificates. While some of these recommendations may have merit, they are beyond the scope of Notice 70-21. However, the FAA is now in the process of investigating and evaluating these recommendations to determine which merit further rule-making action.

One commentator suggested that final action on Notice 70-21 should be deferred until the current FAA/NASA research on aircraft exterior lighting is completed. The FAA does not agree. The changes to the anticollision light requirements proposed in Notice 70-21 are very limited in scope and they are not being dealt with directly in the referenced research. The additional rulemaking that may be generated by the FAA/NASA research would be concerned with the entire area of aircraft exterior lighting and no useful purpose would be served in further delaying the proposed changes.

Several commentators expressed the view that the display of white condenser-discharge anticollision lights significantly reduces the frequency of bird strikes in flight. One respondent submitted statistical data in support of its position and urged that the display of white condenser-discharge lights be made mandatory. The FAA does not agree with these comments. After a thorough review of the data presented, the FAA does not believe that a significant correlation between display of white condenser-discharge lights and frequency of bird strikes has been shown. However, the FAA is continuing its investigation in this area and if further studies reveal that a correlation does exist, it will be the subject of future rule-making action.

Various comments pointed out that while the notice proposed to require an "approved" anticollision light system, it did not specify the standards for such approval. This appears to be a problem

only with respect to anticollision light systems that would have to be installed on aircraft type certificated under airworthiness standards which did not incorporate anticollision light standards. For large aircraft and for certain small aircraft, the operation rules have long required anticollision lights and the airworthiness regulations have long contained standards for such lights. Thus, lights meeting the applicable airworthiness standards are approved lights. In addition, numerous airplanes have, in the past, had anticollision light systems installed voluntarily. All of these systems were approved by the FAA and they continue to be approved. The question as to the identification of approved anticollision light systems arises with respect to the installation of anticollision lights required by this amendment to be installed on aircraft which were certificated under airworthiness standards that did not contain standards for anticollision lights. The FAA is aware that in most instances, the modifications to the older aircraft that would be necessary in order for the anticollision lights to meet the higher intensities proposed in Notice 70-21 would be extensive and may, in certain instances, be prohibitive. This was recognized in the notice and it was not intended that the older aircraft should be required to meet the higher intensities set forth in this amendment. Therefore, the proposal has been revised to make it clear that for the initial installation of anticollision lights on the older aircraft

(those for which type certificates were applied for or issued prior to the effective date of this amendment), the anticollision lights would only need to meet the anticollision light standards in Parts 23, 25, 27, or 29, as applicable, effective immediately prior to the effective date of the amendments contained herein.

Another comment expressed an objection to the proposed increase in the intensity level of anticollision light systems for future aircraft on the grounds that for small aircraft using red anticollision lights, power requirements would be unreasonable, service life short and reliability low, and that for small aircraft using the white anticollision lights, it would not be possible to shield them for purposes of back-scatter without a reduction in the required field-of-coverage. The FAA is aware that for red anticollision lights more electrical power would be needed to meet the new requirements than has been provided in the past. However, the FAA believes that this additional power capacity can be provided on future aircraft at reasonable cost, without incurring a low-service-life or low-reliability penalty. Moreover, a manufacturer would now have the option of installing a white anticollision light,

(As published in the *Federal Register* /36 F.R. 12970/ on July 10, 1971)

thereby eliminating the power problem. The back-scatter problems referred to by the commentator can be solved without diminishing the field-of-coverage by installing a system consisting of three lights, one at each wing-tip and one on the tail.

Finally, one commentator urged the FAA to specify, as part of the revision to the intensity standards for anticollision lights, a minimum infrared signal content for use with PWI systems now under development. This matter was discussed in some detail in Notice 70-21 and the FAA maintains its view that it would be premature to require a minimum infrared signal content until current evaluations by the FAA of the PWI system concept on civil aircraft have been completed. This does not, however, prohibit the incorporation of infrared signal content in any anticollision light system.

In consideration of the foregoing, Parts 23, 25, 27, 29, and 91 of the Federal Aviation Regulations are amended as follows, effective August 11, 1971:

PART 23—AIRWORTHINESS STANDARDS: NORMAL, UTILITY, AND ACROBATIC CATEGORY AIRPLANES

1. Paragraph (c) of § 23.1397 is amended to read as follows:

§ 23.1397 Color specifications.

(c) Aviation white—

"x" is not less than 0.300 and not greater than 0.540;

"y" is not less than "x-0.040" or "y₀-0.010", whichever is the smaller; and

"y" is not greater than "x+0.020" nor "0.636-0.400x";

Where "y₀" is the "y" coordinate of the Planckian radiator for the value of "x" considered.

2. Paragraphs (d) and (f) of § 23.1401 are revised to read as set forth below and paragraph (e) is amended by adding the parenthetical phrase "(if used)" after the word "filter":

§ 23.1401 Anticollision light system.

(d) *Color.* Each anticollision light must be either aviation red or aviation white and must meet the applicable requirements of § 23.1397.

(f) *Minimum effective intensities for anticollision lights.* Each anticollision light effective intensity must equal or exceed the applicable values in the following table.

Angle above or below the horizontal plane:	Effective intensity (candles)
0° to 5°	400
5° to 10°	240
10° to 20°	80
20° to 30°	40

PART 25—AIRWORTHINESS STANDARDS: TRANSPORT CATEGORY AIRPLANES

1. Paragraph (c) of § 25.1397 is amended to read as follows:

§ 25.1397 Color specifications.

(c) Aviation white—

"x" is not less than 0.300 and not greater than 0.540;

"y" is not less than "x-0.040" or "y₀-0.010", whichever is the smaller; and
"y" is not greater than "x+0.020" nor "0.636-0.400x";

Where "y₀" is the "y" coordinate of the Planckian radiator for the value of "x" considered.

2. Paragraphs (d) and (f) of § 25.1401 are revised to read as set forth below and paragraph (e) is amended by adding the parenthetical phrase "(if used)" after the word "filter":

§ 25.1401 Anticollision light system.

(d) *Color.* Each anticollision light must be either aviation red or aviation white and must meet the applicable requirements of § 25.1397.

(f) *Minimum effective intensities for anticollision lights.* Each anticollision light effective intensity must equal or exceed the applicable values in the following table.

Angle above or below the horizontal plane:	Effective intensity (candles)
0° to 5°	400
5° to 10°	240
10° to 20°	80
20° to 30°	40

PART 27—AIRWORTHINESS STANDARDS: NORMAL CATEGORY ROTORCRAFT

1. Paragraph (c) of § 27.1397 is amended to read as follows:

§ 27.1397 Color specifications.

(c) Aviation white—

"x" is not less than 0.300 and not greater than 0.540;

"y" is not less than "x-0.040" or "y₀-0.010", whichever is the smaller; and

"y" is not greater than "x+0.020" nor "0.636-0.400x";

Where "y₀" is the "y" coordinate of the Planckian radiator for the value of "x" considered.

2. Paragraphs (d) and (f) of § 27.1401 are revised to read as set forth below and paragraph (e) is amended by adding the parenthetical phrase "(if used)" after the word "filter":

§ 27.1401 Anticollision light system.

(d) *Color.* Each anticollision light must be either aviation red or aviation white and must meet the applicable requirements of § 27.1397.

(f) *Minimum effective intensities for anticollision light.* Each anticollision light effective intensity must equal or exceed the applicable values in the following table:

Angle above or below the horizontal plane:	Effective intensity (candles)
0° to 5°	400
5° to 10°	240
10° to 20°	80
20° to 30°	40

PART 29—AIRWORTHINESS STANDARDS: TRANSPORT CATEGORY ROTORCRAFT

1. Paragraph (c) of § 29.1397 is amended to read as follows:

§ 29.1397 Color specifications.

(c) Aviation white—

"x" is not less than 0.300 and not greater than 0.540;

"y" is not less than "x-0.040" or "y₀-0.010", whichever is the smaller; and

"y" is not greater than "x+0.020" nor "0.636-0.400x";

Where "y₀" is the "y" coordinate of the Planckian radiator for the value of "x" considered.

2. Paragraphs (d) and (f) of § 29.1401 are revised to read as set forth below and paragraph (e) is amended by adding the parenthetical phrase "(if used)" after the word "filter":

§ 29.1401 Anticollision light system.

(d) *Color.* Each anticollision light must be either aviation red or aviation white and must meet the applicable requirements of § 29.1397.

(f) *Minimum effective intensities for anticollision lights.* Each anticollision light effective intensity must equal or exceed the applicable values in the following table.

Angle above or below the horizontal plane:	Effective intensity (candles)
0° to 5°	400
5° to 10°	240
10° to 20°	80
20° to 30°	40

PART 91—GENERAL OPERATING AND FLIGHT RULES

5. Subparagraph (3) of paragraph (c) of § 91.33 is amended to read as follows:

§ 91.33 Powered civil aircraft with standard category U.S. airworthiness certificates; instrument and equipment requirements.

(c) Visual flight rules (night).

(3) An approved aviation red or aviation white anticollision light system on all large aircraft, on all small aircraft when required by the aircraft's airworthiness certificate, and on all other small aircraft after August 11, 1972. Anticollision light systems initially installed after August 11, 1971, on aircraft for which a type certificate was issued or applied for before August 11, 1971, must at least meet the anticollision light standards of Parts 23, 25, 27, or 29, as applicable, that were in effect on August 10, 1971, except that the color may be either aviation red or aviation white. In the event of failure of any light of the anticollision light system, operations with the aircraft may be continued to a stop where repairs or replacement can be made.

(Secs. 313(a), 601, 603, 604, 49 U.S.C. 1354, 1421, 1423, 1424; sec. 6(c), Department of Transportation Act, 49 U.S.C. 1655(c))

Issued in Washington, D.C., on July 1, 1971.

J. H. SHAFFER,
Administrator.